1. A first-order system with a time constant (τ) has a response that reaches approximately _____ of its final value in 3τ.

   a) 63.2% b) 86.5% c) 95% d) 99%

2. The steady-state error of a type 0 control system to a unit step input is:

   a) Zero b) Infinite c) Non-zero but finite d) Unpredictable

3. The transfer function of a second-order system with no zeros is given by G(s) = 10/(s^2 + 4s + 25). The natural frequency (ω_n) and damping ratio (ζ) of the system are:

   a) ω_n = 4, ζ = 0.25 b) ω_n = 4, ζ = 0.5 c) ω_n = 5, ζ = 0.2 d) ω_n = 5, ζ = 0.4

4. The percent overshoot (%OS) and settling time (t_s) of a second-order system can be determined from its:

   a) Transfer function b) Step response c) Bode plot d) Nyquist plot

5. The poles of a second-order system with a transfer function G(s) = K/[(s + 1)(s + 2)] are located at:

   a) s = -1, -2 b) s = -1 ± j2 c) s = -1/2 ± j(√3)/2 d) s = -1/2, -2

6. The Routh-Hurwitz criterion is used to determine the _____ of a system.

   a) Steady-state error b) Stability c) Gain margin d) Damping ratio

7. In the root locus plot, the intersection of the root locus with the imaginary axis represents:

   a) The gain crossover frequency b) The phase crossover frequency c) The stability margins d) The closed-loop pole locations

8. A lead compensator is designed to improve the _____ of a control system.

   a) Settling time b) Rise time c) Steady-state error d) Damping ratio

9. A lag compensator is designed to improve the _____ of a control system.

   a) Settling time b) Rise time c) Steady-state error d) Damping ratio

10. The phase margin and gain margin can be determined from the _____ of a control system.

    a) Step response b) Bode plot c) Root locus plot d) Nyquist plot

11. The Nyquist plot is used to analyze the _____ of a control system.

    a) Stability b) Damping ratio c) Natural frequency d) Rise time

12. A phase-lead compensator is designed to increase the _____ of a control system.

    a) Damping ratio b) Settling time c) Rise time d) Bandwidth

13. A phase-lag compensator is designed to decrease the _____ of a control system.

    a) Damping ratio b) Settling time c) Rise time d) Bandwidth

14. A PID controller consists of:

    a) Proportional gain only b) Integral action only c) Derivative action only d) Proportional, integral, and derivative actions

15. A P controller is characterized by:

    a) Steady-state error but no overshoot b) Zero steady-state error but overshoot c) Non-zero steady-state error and overshoot d) Zero steady-state error and no overshoot

16. The root-locus method is used to analyze and design control systems based on the behavior of the _____ of the system.

    a) Poles b) Zeros c) Transfer function d) Bode plot

17. Open-loop control systems are more susceptible to _____ compared to closed-loop control systems.

    a) Stability issues b) Steady-state error c) Sensitivity to disturbances d) Oscillations

18. A control system that maintains a desired output by comparing it with the input and adjusting the control action is known as a _____ control system.

    a) Closed-loop b) Open-loop c) Feedback d) Feedforward

19. The transfer function represents the relationship between the _____ of a control system.

a) Input and output signals b) Time response and stability c) Gain and phase margins d) Rise time and settling time

20. The structure diagram of a control system represents the _____ of its components.

a) Connections and interrelationships b) Block diagram representation c) Transfer function coefficients d) Time constants and damping ratios

21. Signal-flow graphs are used to analyze the _____ of a control system.

a) Stability b) Frequency response c) Transfer function d) Signal flow within the system

22. Lead compensators can be implemented in analog control systems using:

a) Resistors and capacitors b) Op-amps and resistors c) Inductors and capacitors d) Diodes and transistors

23. Lag compensators can be implemented in analog control systems using:

a) Resistors and capacitors b) Op-amps and resistors c) Inductors and capacitors d) Diodes and transistors

24. The Bode diagram provides information about the _____ of a control system.

a) Steady-state error b) Frequency response c) Stability margins d) Time response

25. The frequency-domain specifications of a control system include:

a) Rise time and settling time b) Natural frequency and damping ratio c) Gain margin and phase margin d) Overshoot and undershoot

26. Nyquist plots are useful for determining the _____ of a control system.

a) Gain crossover frequency b) Phase crossover frequency c) Stability d) Steady-state error

27. A phase-lead compensator provides a phase shift _____ the crossover frequency.

a) Greater than b) Less than c) Equal to d) Unrelated to

28. A phase-lag compensator provides a phase shift _____ the crossover frequency.

a) Greater than b) Less than c) Equal to d) Unrelated to

29. The PID controller combines proportional, integral, and derivative actions to improve the control system's:

a) Steady-state error b) Stability c) Transient response d) Rise time

30. A PI controller eliminates the steady-state error in response to a _____ input.

a) Step b) Ramp c) Parabolic d) Sinusoidal

Numerical Questions:

1. Calculate the time constant (τ) of a first-order system with a transfer function G(s) = 8/(s + 4).
2. Determine the damping ratio (ζ) and natural frequency (ω_n) of a second-order system with a transfer function G(s) = 6/(s^2 + 3s + 9).
3. Find the steady-state error for a unit step input in a type 1 control system with a transfer function G(s) = 5/s^2.
4. Given a second-order system with a transfer function G(s) = 10/(s^2 + 6s + 25), calculate the percent overshoot (%OS) and settling time (t_s).
5. Calculate the gain K for which the root locus intersects the imaginary axis at s = ±j2 for a transfer function G(s) = K/(s^2 + 4s + 5).
6. Determine the stability of a system with the characteristic equation s^3 + 2s^2 + 4s + 8 = 0 using the Routh-Hurwitz criterion.
7. Design a lead compensator with a transfer function Gc(s) = (s + 0.5)/(s + 2) for a system with a transfer function G(s) = 10/(s + 3).
8. Sketch the Nyquist plot for a system with a transfer function G(s) = 1/(s^2 + 2s + 4).
9. Find the gain crossover frequency and phase crossover frequency from the Bode plot of a system with a transfer function G(s) = 100/(s + 1)(s + 5).
10. Design a PID controller for a system with a transfer function G(s) = 4/(s^2 + 2s + 10).

Multiple-Choice Questions:

1. A phase-lead compensator is used to:

   a) Decrease steady-state error b) Increase system stability c) Improve transient response d) Reduce overshoot

2. The Nyquist stability criterion states that a system is stable if:

   a) The Nyquist plot encircles the (-1, 0) point in the clockwise direction b) The Nyquist plot encircles the (-1, 0) point in the counterclockwise direction c) The Nyquist plot does not encircle the (-1, 0) point d) The Nyquist plot crosses the real axis at least once

3. A PD controller stands for:

   a) Proportional-Differential controller b) Proportional-Derivative controller c) Proportional-Derivative-Integral controller d) Proportional-Differential-Integral controller

4. The root locus technique is used to analyze and design control systems based on:

   a) Bode plots b) Transfer function coefficients c) Pole-zero locations d) Steady-state error

5. A lead compensator:

   a) Increases the gain at high frequencies b) Increases the gain at low frequencies c) Decreases the gain at high frequencies d) Decreases the gain at low frequencies

6. The phase margin is a measure of:

   a) System stability b) Steady-state error c) Transient response d) Control effort

7. The transfer function of a second-order system with no zeros is given by G(s) = 10/(s^2 + 6s + 9). The damping ratio (ζ) of the system is:

   a) 0.5 b) 1 c) 1.5 d) 2

8. The Routh-Hurwitz criterion can determine the stability of a system based on:

   a) The transfer function b) The open-loop gain c) The characteristic equation d) The step response

9. In the root locus plot, the branches start at the _____ and end at the _____.

   a) Poles, zeros b) Zeros, poles c) Zeros, origin d) Poles, origin

10. A phase-lag compensator is designed to improve the _____ of a control system.

    a) Settling time b) Rise time c) Steady-state error d) Damping ratio

1. True/False: A first-order system has only one pole and no zeros. Answer: True
2. True/False: The settling time of a system is the time it takes for the response to reach and stay within a specified error band. Answer: True
3. True/False: The Routh-Hurwitz criterion can determine both stability and performance characteristics of a control system. Answer: False
4. True/False: The phase margin indicates the maximum phase lag that a system can tolerate before becoming unstable. Answer: False
5. True/False: The gain margin provides a measure of the system's stability robustness with respect to parameter variations. Answer: True
6. True/False: A higher gain crossover frequency corresponds to a faster system response. Answer: True
7. True/False: The Nyquist plot provides information about the stability and frequency response of a control system. Answer: True
8. True/False: The phase-lead compensator increases the phase lag at high frequencies. Answer: False
9. True/False: A lag compensator can increase the steady-state error of a system. Answer: ~~True~~
10. True/False: The PID controller combines proportional, integral, and derivative actions to improve system performance. Answer: True